CURRENT ONE-STAGE AND TWO-STAGE IPDE DESIGN WITH EFFICACY

Technical design specification

Purpose

This document describes the current efficacy-enabled IPDE framework: a toxicity CRM with a random carryover parameter, a dose-specific efficacy model with dose-specific carryover, one-stage and two-stage allocation rules, two enrollment schemes, continuous toxicity and futility monitoring, IPDE eligibility, utility definitions, and final OBD selection.

Design family	AIDE/IPDE with binary toxicity and binary efficacy
Dose levels	$j = 1, \ldots, J$
Trial structures	One-stage and two-stage
Enrollment schemes	Continuous enrollment or IPDE-first recruitment
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Design Summary

Core principle. Toxicity determines safety, admissibility, and the allowable direction of dose movement. Efficacy or utility ranks doses only after the toxicity restrictions have been applied. An IPDE treatment is allowed only when both its individual toxicity-risk criterion and its expected efficacy-improvement criterion are satisfied.

The trial structure and enrollment mechanism are separate design choices. Either enrollment scheme may be paired with either the one-stage or the two-stage allocation design.

Component	Option	Main rule
Trial structure	Two-stage	Stage I follows toxicity-based AIDE until `s1_Max` is reached and the next toxicity recommendation is **stay**; Stage II assigns cohorts to the admissible dose with the highest estimated efficacy.
Trial structure	One-stage	The toxicity recommendation is obtained first; a local candidate set is then ranked by utility unless direct de-escalation is required.
Enrollment	Continuous	Enrollment remains open, and newly enrolled patients receive priority when a new cohort is formed.
Enrollment	IPDE-first	Eligible IPDE patients are placed first, and only the remaining cohort positions trigger new-patient recruitment.

Notation

Symbol	Definition	Symbol	Definition
d	Current trial dose	j	Generic dose index
$p_{T,j}$	Toxicity probability for a regular patient at dose j	$p_{E,j}$	Efficacy probability for a regular patient at dose j
r	Global toxicity carryover parameter	r_j	Efficacy carryover parameter for destination dose j
$q_{T,j}$	IPDE toxicity probability at dose j	$q_{E,j}$	IPDE efficacy probability at dose j
ϕ_T	Target toxicity probability	ϕ_E	Minimum acceptable efficacy used for futility
c_T	Posterior toxicity cutoff	c_E	Posterior efficacy-futility cutoff
δ	Minimum efficacy improvement required for IPDE	m_U	User-specified sample threshold in the one-stage rule

1. Statistical Model

The model distinguishes the dose-level probabilities for regular patients from the outcome probabilities for recycled IPDE patients. Toxicity uses a single global carryover parameter r, whereas efficacy uses a dose-specific carryover parameter r_j for each destination dose.

1.1 Toxicity: CRM with a Random Carryover Parameter

Let p_{0j} denote the prespecified CRM skeleton probability at dose j, and let β be the one-dimensional CRM parameter. Under a power CRM,

$$p_{T,j}(\beta) = p_{0j}^{\exp(\beta)}, \qquad \beta \sim N(\mu_\beta, \sigma_\beta^2).$$

For a regular patient treated at dose j,

$$Y_{T,j}^{(R)} \mid \beta \sim \text{Bernoulli}(p_{T,j}) \,.$$

For an IPDE patient recycled to dose j,

$$Y_{T,j}^{(I)} \mid \beta, r \sim \text{Bernoulli}(q_{T,j}) \,, \qquad q_{T,j} = r + (1-r)p_{T,j}, \qquad r \sim \text{Beta}(a_r, b_r).$$

The toxicity carryover parameter is global: IPDE toxicity observations from all dose levels contribute to the posterior of r. When $r = 0$, the IPDE toxicity probability reduces to the regular-patient CRM probability. As r increases, $q_{T,j}$ is shifted upward toward one.

The posterior distribution of (β, r) is updated using all available regular-patient and IPDE toxicity outcomes. Dose-level toxicity decisions use posterior summaries of $p_{T,j}$, while patient-specific IPDE risk assessment uses $q_{T,j}$.

1.2 Efficacy: Dose-Specific Beta-Binomial Model

The regular-patient efficacy probability is modeled separately at each dose:

$$p_{E,j} \sim \text{Beta}(a_{E,j}, b_{E,j}), \qquad j = 1, \ldots, J.$$

For a regular patient at dose j,

$$Y_{E,j}^{(R)} \mid p_{E,j} \sim \text{Bernoulli}(p_{E,j}) \,.$$

For an IPDE patient recycled to dose j,

$$Y_{E,j}^{(I)} \mid p_{E,j}, r_j \sim \text{Bernoulli}(q_{E,j}) \,, \qquad q_{E,j} = r_j + (1-r_j)p_{E,j}, \qquad r_j \sim \text{Beta}(a_{r,j}, b_{r,j}).$$

Each destination dose has its own r_j. Thus, the carryover effect may vary by dose rather than being forced to share a common value. The current specification is non-hierarchical: the priors for $p_{E,j}$ and r_j are prespecified independently for each dose.

The baseline probability $p_{E,j}$ is used for ordinary dose comparison and utility calculation. The IPDE probability $q_{E,j}$ is used when evaluating the expected efficacy of recycling a particular patient to dose j.

1.3 Grouped Likelihood Contributions

Let $n_{T,j}^{(R)}$ and $y_{T,j}^{(R)}$ be the number of regular patients and DLTs at dose j, and define $n_{T,j}^{(I)}$ and $y_{T,j}^{(I)}$ analogously for IPDE patients. The toxicity likelihood is

$$L_T(\beta, r) \propto \prod_{j=1}^{J} p_{T,j}^{y_{T,j}^{(R)}} (1 - p_{T,j})^{n_{T,j}^{(R)} - y_{T,j}^{(R)}} \times q_{T,j}^{y_{T,j}^{(I)}} (1 - q_{T,j})^{n_{T,j}^{(I)} - y_{T,j}^{(I)}}.$$

The efficacy likelihood is

$$L_E(\{p_{E,j}, r_j\}_{j=1}^{J}) \propto \prod_{j=1}^{J} p_{E,j}^{y_{E,j}^{(R)}} (1 - p_{E,j})^{n_{E,j}^{(R)} - y_{E,j}^{(R)}} \times q_{E,j}^{y_{E,j}^{(I)}} (1 - q_{E,j})^{n_{E,j}^{(I)} - y_{E,j}^{(I)}}.$$

Decision summary convention. Posterior means are the default point estimates in the rules below. Posterior medians may be used instead, but one summary convention should be applied consistently across all doses and all decision times.

2. IPDE Trial Design

At every decision time, safety is evaluated before efficacy or utility. Doses eliminated for overtoxicity or futility are removed from cohort-allocation and IPDE-destination sets.

2.1 Two-Stage Design

1. **Stage I: toxicity-driven AIDE.** Patients are treated according to the toxicity-based AIDE design until the cumulative Stage I sample size has reached the user-specified limit `s1_Max` and the next toxicity recommendation is **stay**. Efficacy outcomes are collected and the efficacy model is updated, but Stage I cohort assignments are determined by toxicity.

2. **Continuous monitoring during Stage I.** The overtoxicity, futility, and IPDE rules in Section 3 are evaluated whenever the posterior distributions are updated. Thus, an efficacy-futile dose may be eliminated during Stage I even though efficacy does not drive the Stage I cohort assignment.

3. **Stage II: efficacy-driven allocation.** Among the currently admissible doses, assign the next cohort to the dose with the largest posterior efficacy estimate, ordinarily

$$\widehat{p}_{E,j} = E(p_{E,j} \mid \mathcal{D}).$$

   A dose cannot be selected if it has been eliminated for overtoxicity or futility.

4. **Stage II recycling.** Each potentially recyclable patient is assessed against the destination-specific criteria in Section 3.3. If `flexible = FALSE`, the permitted destination must be below the current trial dose. If `flexible = TRUE`, any admissible dose may be considered, including a higher dose.

Stage II ranking set. The efficacy comparison is made only among admissible doses. Any tried-dose restriction, dose-skipping restriction, or other implementation constraint is applied before identifying the dose with the highest estimated efficacy.

2.2 One-Stage Design

Let d denote the current dose, n_d the number of toxicity-evaluable patients at d, and m_U the user-specified threshold for activating the reduced local utility set. The current working example is $m_U = 6$, but this value is not fixed by the method.

The toxicity recommendation is obtained first.

Toxicity decision	Condition	Cohort-allocation rule
De-escalate	Any n_d	De-escalate according to the toxicity rule. Utility is not permitted to override a direct safety-driven de-escalation.
Stay	$n_d \geq m_U$	Rank the admissible doses in $\{d-1, d\}$ by utility and assign the cohort to the dose with the largest utility.
Escalate or stay	Escalate, or stay with $n_d < m_U$	Rank the admissible doses in $\{d-1, d, d+1\}$ by utility and assign the cohort to the dose with the largest utility.

Before utility ranking, candidate indices are truncated to $\{1, \ldots, J\}$, and overtoxic, futile, or otherwise prohibited doses are removed. If only one candidate remains, that dose is assigned without a utility comparison. Under a TITE implementation, n_d may be replaced by the corresponding toxicity effective sample size if that is how the current software defines the threshold.

2.3 Enrollment Schemes

Scheme	Queue priority	Operational rule
Continuous enrollment	New patients first	Enrollment remains open. When a new cohort is formed, newly enrolled patients are placed first, and eligible IPDE patients fill the remaining positions. Patients arriving while a cohort is closed wait in the appropriate queue.
IPDE-first recruitment	Eligible IPDE patients first	When a cohort is ready to form, place eligible IPDE patients first. Recruit only the number of new patients needed to complete the cohort. New-patient enrollment is therefore activated as needed rather than continuously.

Cohort formation. A cohort opens only when the design is ready to make a new assignment. The enrollment scheme determines who fills the available positions; the one-stage or two-stage design determines the assigned dose.

3. Overtoxicity, Futility, and IPDE Criteria

The following checks are repeated throughout the trial whenever outcomes become available and the posterior distributions are updated.

3.1 Overtoxicity Monitoring

For each dose j, eliminate the dose for overtoxicity when

$$\text{Pr}(p_{T,j} > \phi_T \mid \mathcal{D}) > c_T.$$

An eliminated dose cannot receive a new cohort and cannot be used as an IPDE destination. Under the monotonicity assumption for toxicity, elimination of dose j may also eliminate all higher doses. The trial terminates early when no admissible safe dose remains; in particular, this occurs when the lowest dose is eliminated by the overtoxicity rule.

3.2 Efficacy Futility Monitoring

Eliminate dose j for futility when

$$\text{Pr}(p_{E,j} < \phi_E \mid \mathcal{D}) > c_E.$$

A futile dose cannot receive a new cohort, cannot be selected as an IPDE destination, and is excluded from final OBD selection. Futility monitoring remains active in both stages of the two-stage design and throughout the one-stage design.

3.3 IPDE Eligibility and Destination Criteria

Suppose a patient was previously treated at dose ℓ and is being considered for recycling to destination dose k. The patient may be recycled to k only when all five conditions are satisfied.

1. **Individual toxicity risk.**

$$\text{Pr}(q_{T,k} > \phi_T \mid \mathcal{D}) < c_T, \qquad q_{T,k} = r + (1-r)p_{T,k}.$$

   The posterior probability that the patient's IPDE toxicity risk exceeds the target must remain below the prespecified cutoff.

2. **Expected efficacy improvement.**

$$E(q_{E,k} - p_{E,\ell} \mid \mathcal{D}) > \delta, \qquad q_{E,k} = r_k + (1-r_k)p_{E,k}.$$

   Under the default plug-in implementation, this is evaluated as

$$\widehat{r}_k + (1-\widehat{r}_k)\widehat{p}_{E,k} - \widehat{p}_{E,\ell} > \delta.$$

3. **Cycle limit.** The patient satisfies

$$n_{\text{cycle}} < \texttt{max\_cycle}.$$

4. **Observed outcomes.** The patient has experienced neither a DLT nor an efficacy response during the completed cycle used to determine recycling eligibility.

5. **Dose-direction restriction.** If `flexible = FALSE`, the destination dose must be below the current trial dose. If `flexible = TRUE`, any admissible destination may be considered, including a higher dose.

If no destination satisfies all five conditions, the patient is not recycled under the IPDE rule.

3.4 Order of Operations

1. Update the posterior of the CRM parameter β, the toxicity carryover r, each efficacy probability $p_{E,j}$, and each efficacy carryover r_j.
2. Apply the dose-level overtoxicity and futility rules and update the admissible dose set.
3. Determine the next cohort dose using the active one-stage or two-stage allocation rule.
4. Evaluate each potential IPDE patient against the five patient- and destination-level criteria before cohort placement.

4. Utility Functions

Utilities are based on the baseline dose-level probabilities $p_{E,j}$ and $p_{T,j}$. The carryover parameters affect their posterior estimation through the IPDE likelihood contributions, but the patient-specific probabilities $q_{E,j}$ and $q_{T,j}$ are reserved for recycling decisions.

4.1 Utility 1: Efficacy Only

$$U_1(j) = p_{E,j}.$$

This utility ranks admissible doses solely by estimated efficacy. Toxicity affects the candidate set through the safety rules and the MTD ceiling, rather than through the numerical utility value.

4.2 Utility 2: Efficacy Minus a Toxicity Penalty

$$U_2(j) = p_{E,j} - \lambda_T p_{T,j}, \qquad \lambda_T \geq 0.$$

The prespecified parameter λ_T controls the toxicity penalty. A larger value places greater weight on avoiding toxicity. The notation λ_T is used to avoid confusion with the CRM parameter β and any carryover parameter.

4.3 Utility 3: Four-Outcome Joint Score

Assign a score u_{te} to each joint outcome $(T = t, E = e)$, where $t, e \in \{0, 1\}$. Under the current working-independence approximation, the joint probabilities are products of the toxicity and efficacy marginals.

	$E = 0$	$E = 1$
$T = 0$	u_{00} with probability $(1 - p_{T,j})(1 - p_{E,j})$	u_{01} with probability $(1 - p_{T,j})p_{E,j}$
$T = 1$	u_{10} with probability $p_{T,j}(1 - p_{E,j})$	u_{11} with probability $p_{T,j}p_{E,j}$

The expected utility is

$$\begin{aligned} U_3(j) = {} & u_{00}(1 - p_{T,j})(1 - p_{E,j}) + u_{01}(1 - p_{T,j})p_{E,j} \\ & + u_{10}p_{T,j}(1 - p_{E,j}) + u_{11}p_{T,j}p_{E,j}. \end{aligned}$$

Working independence. The product form does not estimate a toxicity-efficacy association parameter. A joint model may replace these products later without changing the surrounding allocation framework.

Only admissible doses are ranked using a common posterior summary. Use a prespecified tie-breaking rule; selecting the lower dose is a conservative default. Utility cannot override a direct toxicity-driven de-escalation.

5. Final OBD Selection

Final selection is sequential: the toxicity model determines the MTD ceiling, the efficacy model removes futile doses, and utility identifies the optimal dose among the remaining candidates.

1. **Select the MTD from the toxicity model.** Apply the implemented CRM MTD rule using posterior estimates of $p_{T,j}$. By default, restrict selection to tried doses and exclude doses eliminated for overtoxicity. A standard CRM rule selects the tried admissible dose whose posterior toxicity estimate is closest to ϕ_T.
2. **Remove efficacy-futile doses.** Exclude every dose satisfying

$$\Pr(p_{E,j} < \phi_E \mid \mathcal{D}) > c_E.$$

3. **Construct the OBD candidate set.** Under the default interpretation that "below the MTD" means "not higher than the MTD," define

$$\mathcal{C}_{\text{OBD}} = \left\{ j : j \text{ has been tried}, \; j \le \widehat{j}_{\text{MTD}}, \; j \text{ is neither overtoxic nor futile} \right\}.$$

4. **Select the highest-utility dose.**

$$\widehat{j}_{\text{OBD}} = \operatorname*{arg\,max}_{j \in \mathcal{C}_{\text{OBD}}} U(j),$$

where U is Utility 1, 2, or 3.

If $\mathcal{C}_{\text{OBD}}$ is empty, no OBD is selected. The design should not force an OBD when all tried doses are overtoxic, efficacy-futile, or otherwise inadmissible.

Default Restrictions and Reporting

- MTD selection is restricted to tried doses by default.
- OBD selection is restricted to tried doses by default.
- Stage II efficacy ranking and one-stage utility ranking may also apply tried-dose or adjacency restrictions, depending on the implementation; these restrictions should be stated in simulation reports.
- Report the selected MTD and OBD, posterior toxicity and efficacy estimates, selected utility, dose eliminations, number and destinations of IPDE treatments, trial duration, regular-patient enrollment, and no-selection outcomes.

Compact Algorithm Summary

Two-Stage Algorithm

1. Initialize the toxicity and efficacy models at the starting dose.
2. During Stage I, allocate cohorts by toxicity-based AIDE until `s1_Max` is reached and the next toxicity recommendation is **stay**.
3. At every update, apply overtoxicity, futility, and IPDE checks.
4. During Stage II, assign cohorts to the admissible dose with the largest posterior efficacy estimate.
5. At the final analysis, select the MTD, remove futile doses, and select the highest-utility tried dose not exceeding the MTD.

One-Stage Algorithm

1. Update the toxicity CRM at the current dose d and obtain the toxicity recommendation.
2. If the recommendation is de-escalation, de-escalate directly.

3. If the recommendation is stay and $n_d \geq m_U$, rank $\{d-1, d\}$ by utility after exclusions.
4. If the recommendation is escalation, or stay with $n_d < m_U$, rank $\{d-1, d, d+1\}$ by utility after exclusions.
5. Evaluate potential IPDE patients separately using the destination-specific criteria before cohort placement.

User-Specified Control Parameters

Parameter	Role	Convention
`s1_Max`	Stage I sample-size limit	User specified
`m_U`	One-stage stay-rule threshold	User specified; current example $m_U = 6$
`phi_T`	Target toxicity probability	User specified
`c_T`	Toxicity posterior cutoff	User specified
`phi_E`	Minimum acceptable efficacy	User specified
`c_E`	Efficacy-futility posterior cutoff	User specified
`delta`	Minimum efficacy improvement for IPDE	User specified
`max_cycle`	Maximum cycles per patient	User specified
`flexible`	Permitted IPDE destination range	FALSE: below current dose; TRUE: any admissible dose
`enrollment_scheme`	Cohort queue mechanism	Continuous new-patient priority or IPDE-first recruitment
`utility_type`	Dose-ranking function	1, 2, or 3
`lambda_T`	Toxicity penalty in Utility 2	User specified
`tried`	Final-selection restriction	TRUE by default for MTD and OBD

6. Additive Previous-Dose Outcome Models

The current implementation adds the contribution from the immediately preceding dose directly to the regular-patient probability for an IPDE administration. This section applies when the `previous_dose_additive` toxicity and efficacy models are selected and replaces the random carryover specifications in Sections 1.1–1.3 for that analysis.

Let i index an administration, let j_i denote its current dose, let ℓ_i denote the same patient's immediately preceding dose, and let $I_i = 1$ indicate an IPDE administration, with $I_i = 0$ indicating a regular administration.

6.1 Toxicity: Additive Previous-Dose CRM

The regular-patient toxicity probability at dose j remains the power CRM:

$$p_{T,j}(\beta) = p_{0j}^{\exp(\beta)}, \qquad \beta \sim N(\mu_\beta, \sigma_\beta^2).$$

For a regular administration,

$$Y_{T,i} \mid \beta, I_i = 0 \sim \text{Bernoulli}\left(p_{T,j_i}(\beta)\right).$$

For an IPDE administration,

$$Y_{T,i} \mid \beta, \alpha_T, I_i = 1 \sim \text{Bernoulli}(q_{T,i}), \qquad q_{T,i} = \min\left\{1,\ p_{T,j_i}(\beta) + \alpha_T p_{T,\ell_i}(\beta)\right\},$$
$$\alpha_T \sim \text{Beta}(a_{\alpha_T}, b_{\alpha_T}).$$

The coefficient α_T is shared across IPDE toxicity administrations. When $\alpha_T = 0$, the IPDE toxicity probability equals the regular toxicity probability at the current dose. Larger values add a greater fraction of the regular toxicity risk associated with the patient's preceding dose, with the resulting probability capped at one.

6.2 Efficacy: Additive Previous-Dose Beta-Binomial Model

The regular-patient efficacy probability is modeled separately at each dose:

$$p_{E,j} \sim \text{Beta}(a_{E,j}, b_{E,j}), \qquad j = 1, \dots, J.$$

For a regular administration,

$$Y_{E,i} \mid p_{E,j_i}, I_i = 0 \sim \text{Bernoulli}(p_{E,j_i}).$$

For an IPDE administration,

$$Y_{E,i} \mid \{p_{E,j}\}_{j=1}^{J}, \alpha_E, I_i = 1 \sim \text{Bernoulli}(q_{E,i}),$$
$$q_{E,i} = \min\left\{1,\ p_{E,j_i} + \alpha_E p_{E,\ell_i}\right\}, \qquad \alpha_E \sim \text{Beta}(a_{\alpha_E}, b_{\alpha_E}).$$

The efficacy coefficient α_E is distinct from α_T and from the dose-specific Beta prior parameters $a_{E,j}$ and $b_{E,j}$. The efficacy model uses the same current/previous dose pair (j_i, ℓ_i) as the toxicity model rather than assigning a separate carryover parameter to every destination dose.

6.3 Likelihood, Posterior Use, and Operating Rules

Because $q_{T,i}$ and $q_{E,i}$ depend on the patient's actual dose pair (j_i, ℓ_i), the likelihood is evaluated at the individual-administration level. Define

$$\pi_{T,i} = \begin{cases} p_{T,j_i}(\beta), & I_i = 0, \\ q_{T,i}, & I_i = 1, \end{cases} \qquad \pi_{E,i} = \begin{cases} p_{E,j_i}, & I_i = 0, \\ q_{E,i}, & I_i = 1. \end{cases}$$

The toxicity likelihood contribution is

$$L_T(\beta, \alpha_T) \propto \prod_{i=1}^{n_T} \pi_{T,i}^{Y_{T,i}} (1 - \pi_{T,i})^{1-Y_{T,i}}.$$

The efficacy likelihood contribution is

$$L_E\left(\{p_{E,j}\}_{j=1}^{J}, \alpha_E\right) \propto \prod_{i=1}^{n_E} \pi_{E,i}^{Y_{E,i}} (1 - \pi_{E,i})^{1-Y_{E,i}}.$$

This individual-level form is required because two IPDE patients treated at the same destination dose may have different preceding doses and therefore different additive probabilities.

Posterior quantities

- The toxicity fit estimates β, α_T, and the regular-dose probabilities $p_{T,j}$ using all regular and IPDE toxicity outcomes.
- The efficacy fit estimates α_E and every regular-dose probability $p_{E,j}$ using all regular and IPDE efficacy outcomes.
- For a candidate IPDE treatment, $q_{T,i}$ and $q_{E,i}$ are recomputed from posterior draws using that patient's actual pair (ℓ_i, j_i).

Decision interpretation

- Use $p_{T,j}$ for dose-level safety, MTD selection, and utility calculations; use $q_{T,i}$ for the individual IPDE safety criterion.
- Use $p_{E,j}$ for efficacy ranking, futility monitoring, and utility calculations; use $q_{E,i}$ for the expected efficacy improvement required for recycling.
- The one-stage and two-stage allocation rules, enrollment schemes, overtoxicity rule, and futility rule are otherwise unchanged.

Implementation note. The additive coefficients may be fixed or estimated using prespecified Beta priors, and the toxicity and efficacy coefficients may be specified separately. The cap min$\{1, \cdot\}$ is applied to every IPDE probability so that each additive probability remains in $[0, 1]$.